XRPL Validator — Daily Engineering Report · 2026-09-10

Period: 2026-09-10 00:00 to 24:00 US Eastern (2026-09-10 04:00 to 2026-09-11 04:00 UTC); ledgers #106,882,088 to #106,904,495. Every count from the ledger and the validator's logs stops at #106,904,495. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; Step 1 receipt records (they begin 2026-08-29); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** 24 engine receipts recorded (10 fixed, 0 explained, 14 without a recorded fix).
- **Changes:** 0 deployments, 9 commits merged into main, 10 engine findings fixed (F242–F251).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

Ledgers with a receipt in the period (#106,882,088–#106,904,495), by step:

Step	Ledgers with a receipt
Step 1 · native engine	24
Step 2 · records	0
Step 3 · order	0

Step	Ledgers with a receipt
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#106,884,621	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-02	—	#106,889,388	Step 1 · native engine	1 object(s) differed in content	fixed: F248, commit 620104b, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-03	—	#106,891,190	Step 1 · native engine	1 object(s) differed in content	fixed: F249, commit 1613c3b, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-04	—	#106,891,330	Step 1 · native engine	2 object(s) differed in content	fixed: F250, commit c7ce688, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-05	—	#106,891,531	Step 1 · native engine	1 object(s) differed in content	fixed: F250, commit c7ce688, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-06	—	#106,894,782	Step 1 · native engine	1 transaction result(s) differed from rippled's	fixed: F247, commit a3aa092, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-07	—	#106,898,039	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	fixed: F245, commit c38fa03, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-08	—	#106,898,043	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	no disposition recorded
R-09	—	#106,898,047	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	no disposition recorded
R-10	—	#106,898,058	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	no disposition recorded
R-11	—	#106,898,066	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	no disposition recorded
R-12	—	#106,898,071	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	no disposition recorded
R-13	—	#106,898,092	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	no disposition recorded
R-14	—	#106,898,099	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	no disposition recorded

ID	Time	Ledger(s)	Step	Description	Disposition
R-15	—	#106,898,110	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	no disposition recorded
R-16	—	#106,898,114	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	no disposition recorded
R-17	—	#106,898,118	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	no disposition recorded
R-18	—	#106,898,135	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	no disposition recorded
R-19	—	#106,898,237	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	no disposition recorded
R-20	—	#106,898,403	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	no disposition recorded
R-21	—	#106,898,413	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make	fixed: F245, commit c38fa03, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-22	—	#106,900,377	Step 1 · native engine	1 transaction result(s) differed from rippled's; 5 ledger change(s) the network made were absent from ours: 73B2DBB8, 222C60F3, 50850843	fixed: F246, commit 48567ac, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-23	—	#106,900,393	Step 1 · native engine	1 transaction result(s) differed from rippled's; 5 ledger change(s) the network made were absent from ours: 08C1F6F4, 50850843, 222C60F3	fixed: F246, commit 48567ac, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-24	—	#106,900,408	Step 1 · native engine	1 transaction result(s) differed from rippled's; 5 ledger change(s) the network made were absent from ours: F52A477B, 08C1F6F4, 4A871F7F (deleted)	fixed: F246, commit 48567ac, deployed in cycle 121 (2026-09-11 17:39 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-10-01, INC-01 is INC-2026-09-10-01. Times are not shown where the reference node no longer holds the ledger.

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merged	Repo	Commit	Subject
02:55 UTC	xrpl-validator	`a3aa092`	fix(xrpl-ledger): F247 — a two-asset deposit that rounds to zero LP tokens is tecAMM_INVALID_TOKENS
02:55 UTC	xrpl-validator	`48567ac`	fix(xrpl-ledger): F246 — anyone may cancel an EXPIRED check
02:55 UTC	xrpl-validator	`c38fa03`	fix(xrpl-ledger): F245 — accepting an expired NFTokenOffer deletes it and is tecEXPIRED
02:55 UTC	xrpl-validator	`60b4142`	fix(xrpl-ledger): F244 — a bridged crossing's leg-B rev extent is withdrawn: it reaped offers rippled's stream never steps onto
02:55 UTC	xrpl-validator	`3fefb0d`	fix(xrpl-ledger): F242 — unauthorizing a DepositPreauth drops the owner directory it empties
03:47 UTC	xrpl-validator	`620104b`	fix(xrpl-ledger): F248 — a payment hop's limiting book pass re-runs with the level's sixteen-digit fold, like a crossing's
03:47 UTC	xrpl-validator	`1613c3b`	fix(xrpl-ledger): F249 — a bridge member consumed at its own funds cap rounds its in DOWN
03:47 UTC	xrpl-validator	`c7ce688`	fix(xrpl-ledger): F250 — a bridge level's ending member keeps its own ceil_out; the fold's lost digit is the taker's
03:47 UTC	xrpl-validator	`6328e1b`	fix(xrpl-ledger): F243 — a crossing's pool round is capped by the GROSS fold remainder, not the net remainder re-grossed

Findings fixed

- **fix(xrpl-ledger): F247 — a two-asset deposit that rounds to zero LP tokens is tecAMM_INVALID_TOKENS** (`a3aa092` , xrpl-validator) `equalDepositLimit` (AMMDeposit.cpp:735-741, :762-767) reports a lead direction whose `tokensAdj == beast::zero` as tecAMM_INVALID_TOKENS under fixAMMv1_3, judged before that direction's fit and ending the search; only "neither direction fits" is tecAMM_FAILED. Our `equal_deposit_limit` folded both into one None → tecAMM_FAILED. #106894782 ED009182D8E8: 7e-13 589 and 6e-13 PLX into the 589/PLX pool — fee-only either …
- **fix(xrpl-ledger): F246 — anyone may cancel an EXPIRED check** (`48567ac` , xrpl-validator) CancelCheck.cpp:56-66 restricts the cancel to the writer or the destination only while the check is live: `if (!optExpiry || parentCloseTime < Expiration)` guards the tecNO_PERMISSION test, so an expired check is cancellable by any account. We tested the parties unconditionally. #106900377 A252917DB8E8 (then #106900393, #106900408): rpcjmdWFCQ — neither writer nor destination — cancels rDoxEVqFy3's check 3FF6163A to …
- **fix(xrpl-ledger): F245 — accepting an expired NFTokenOffer deletes it and is tecEXPIRED** (`c38fa03` , xrpl-validator) rippled 3.3.0 under fixCleanup3_1_3 (NFTokenAcceptOffer.cpp:69-75, :446-477): an expired offer passes preclaim, and doApply's FIRST act deletes every named offer that has expired — hasExpired(view(), Expiration), i.e. parentCloseTime >= Expiration, buy then sell — and returns tecEXPIRED if any was. The Transactor's processPersistentChanges then re-applies exactly those erasures after the tec reset (ltNFTOKEN_OFFER …
- **fix(xrpl-ledger): F244 — a bridged crossing's leg-B rev extent is withdrawn: it reaped offers rippled's stream never steps onto** (`60b4142` , xrpl-validator) Finding 239 added `rev_extent_reap` to cross_bridged so a bridged strand's rev pass would remove expired/unfunded offers past the level it consumed whole (#106863376). Four ledgers of the deploy109 soak show its reach is wrong: #106873753 C4FEF57142AA — rMsXVzCug7's tfPassive BTC→RLUSD offer — crosses NOTHING (mainnet touches six nodes and rests it), yet the extent walked the BTC→XRP→RLUSD bridge and reaped …

- **fix(xrpl-ledger): F242 — unauthorizing a DepositPreauth drops the owner directory it empties** (`3fefb0d`, xrpl-validator) rippled unlinks a DepositPreauth with keepRoot FALSE: view.dirRemove(keylet::ownerDir(account), page, preauthIndex, false) DepositPreauth.cpp:283
- **fix(xrpl-ledger): F248 — a payment hop's limiting book pass re-runs with the level's sixteen-digit fold, like a crossing's** (`620104b`, xrpl-validator) Finding 149 ported StrandFlow's limiting re-execution for OfferCreate crossings only (`fold_rem`): when a step's rev output is not what the strand asked for, rippled throws the pass away and re-runs that step's rev with its own 16-digit total as the target (StrandFlow.h:176-184), so the level's LAST whole offer is re-sized by limitStepOut to `total16 − Σprev` at the page rate. The rule is the strand's, not the …
- **fix(xrpl-ledger): F249 — a bridge member consumed at its own funds cap rounds its in DOWN** (`1613c3b`, xrpl-validator) rippled's forEachOffer funds branch (BookStep.cpp:779-793): when the owner cannot fund the whole offer, `ownerGives = funds; stpAmt.out = mulRatio(funds, …, roundUp = false); ofrAmt = offer.limitOut(ofrAmt, stpAmt.out, roundUp = false)` — the member's in is the page-rate product rounded DOWN ("so that the quality of an offer left in the ledger is as good or better than its book page"). Only a slice the STRAND cuts …
- **fix(xrpl-ledger): F250 — a bridge level's ending member keeps its own ceil_out; the fold's lost digit is the taker's** (`c7ce688`, xrpl-validator) In an OUT-limited pass rippled sizes the level's ending member with limitStepOut — `offer.limitOut(ofrAmt, out', roundUp = true)`, i.e. its OWN ceil_out at the page rate — and the pass's in is `sum(savedIns)`, the sixteen-digit fold of the members' ins (BookStep.cpp revImp). When that fold drops a seventeenth digit, the taker pays the folded figure and the makers keep theirs. Our dealing plan gave the ending member …
- **fix(xrpl-ledger): F243 — a crossing's pool round is capped by the GROSS fold remainder, not the net remainder re-grossed** (`6328e1b`, xrpl-validator) rippled's strand keeps `remainingIn` gross-primary — `maxIn − sum(savedIns)` at sixteen digits (StrandFlow) — and an iteration that exhausts it debits the taker that remainder verbatim; the net the pool receives is derived from it. cross_bridged handed its pool turns (`consume`/`consume_fib`) a round cap only when the trust line bound tighter (finding 212); otherwise the exhausting slice re-grossed the NET remainder …
- **fix(xrpl-ledger): F251 — a bridged leg's rev extent reaps only after a CLOB-served pass; a pool-served pass never steps onto the book** (`5de9f43`, xrpl-validator) Finding 239 gave leg B of a bridged crossing rippled's rev-pass reach — after consuming a level, the stream steps on, removing the expired and unfunded offers it lands on until the first live one. Finding 244 withdrew it: on #106873753 C4FEF57142AA (rMsXVzCug7's tfPassive BTC→RLUSD offer, which crosses nothing) the extent removed rLDyWWMiW6's dead EE68DEE8 and its pages, four ledgers running, while mainnet left …

Independence program

No independence record for UTC day 2026-09-10; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_3_0	2026-08-28 11:15 UTC	2026-09-11 11:15 UTC

Amendments enabled on mainnet: 93.

Open items

- Receipt R-01 (#106,884,621): open, under investigation.
- Receipt R-08 (#106,898,043): open, under investigation.
- Receipt R-09 (#106,898,047): open, under investigation.
- Receipt R-10 (#106,898,058): open, under investigation.
- Receipt R-11 (#106,898,066): open, under investigation.
- Receipt R-12 (#106,898,071): open, under investigation.
- Receipt R-13 (#106,898,092): open, under investigation.
- Receipt R-14 (#106,898,099): open, under investigation.
- Receipt R-15 (#106,898,110): open, under investigation.
- Receipt R-16 (#106,898,114): open, under investigation.
- Receipt R-17 (#106,898,118): open, under investigation.
- Receipt R-18 (#106,898,135): open, under investigation.
- Receipt R-19 (#106,898,237): open, under investigation.
- Receipt R-20 (#106,898,403): open, under investigation.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **State-Triggered Escrow via Credential Status** (payments) Add a tfStateConditional flag to EscrowCreate plus SFields sfConditionKeylet (Hash256) and sfConditionCode (UInt8 enum: CREDENTIAL_VALID, CREDENTIAL_R
- **Oracle-Bounded Slippage Protection for IOC Offers** (dex) Add a tfProtectedFill flag to OfferCreate plus SFields sfPriceOracleID (a PriceOracle keylet) and sfMaxSlippageBps (UInt16). In the existing offer-cro
- **Validator Fee-Vote Extended to Tunable Parameters** (governance) Add a new ltPARAMS singleton ledger object holding a small fixed whitelist of tunable numeric constants (initially: fee-escalation multiplier, tx-queu

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-11 02:55	09-11 02:55	xrpl-validator	`a3aa092`	fix(xrpl-ledger): F247 — a two-asset deposit that rounds to zero LP tokens is tecAMM_INVALID_TOKENS
09-11 02:55	09-11 02:55	xrpl-validator	`48567ac`	fix(xrpl-ledger): F246 — anyone may cancel an EXPIRED check
09-11 02:55	09-11 02:55	xrpl-validator	`c38fa03`	fix(xrpl-ledger): F245 — accepting an expired NFTokenOffer deletes it and is tecEXPIRED
09-11 02:55	09-11 02:55	xrpl-validator	`60b4142`	fix(xrpl-ledger): F244 — a bridged crossing's leg-B rev extent is withdrawn: it reaped offers rippled's stream never steps onto
09-09 15:03	09-11 02:55	xrpl-validator	`3fefb0d`	fix(xrpl-ledger): F242 — unauthorizing a DepositPreauth drops the owner directory it empties
09-11 03:15	09-11 03:47	xrpl-validator	`620104b`	fix(xrpl-ledger): F248 — a payment hop's limiting book pass re-runs with the level's sixteen-digit fold, like a crossing's
09-11 03:19	09-11 03:47	xrpl-validator	`1613c3b`	fix(xrpl-ledger): F249 — a bridge member consumed at its own funds cap rounds its in DOWN
09-11 03:31	09-11 03:47	xrpl-validator	`c7ce688`	fix(xrpl-ledger): F250 — a bridge level's ending member keeps its own ceil_out; the fold's lost digit is the taker's
09-11 03:43	09-11 03:47	xrpl-validator	`6328e1b`	fix(xrpl-ledger): F243 — a crossing's pool round is capped by the GROSS fold remainder, not the net remainder re-grossed
09-11 03:59	—	xrpl-validator	`5de9f43`	fix(xrpl-ledger): F251 — a bridged leg's rev extent reaps only after a CLOB-served pass; a pool-served pass never steps onto the book

Times in UTC. A commit appears on the day it was committed to main or merged into it.